

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA5116 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)*

Assessment Tool Weightage: 40%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

*I KESHAVARAM L /192511253(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

L. Keshavaram

Annexure A

CONSTRAINT-BASED PROBLEM SOLVING

1. Intrusion Detection Deployment under Real-Time Constraints

A financial organization plans to deploy an Intrusion Detection System (IDS) with the following constraints:

- Must analyze traffic in real-time with latency ≤ 5 ms per packet
- Detection accuracy $\geq 96\%$
- False positive rate $\leq 1.5\%$
- Network traffic load: 150,000 packets per second
- Limited CPU resources (shared with other services)

Task:

Design an optimized IDS configuration (signature-based, anomaly-based, or hybrid). Analyze trade-offs between detection accuracy and system performance, and justify how your solution meets all constraints.

2. Worm Containment Strategy under Network Availability Constraints

An enterprise network faces rapid worm propagation with the following constraints:

- Network downtime must be $\leq 1\%$
- Detection and containment must occur within 2 minutes
- Systems are distributed across multiple locations
- Limited centralized control

Task:

Design a containment strategy using firewalls, IDS/IPS, and segmentation. Evaluate trade-offs between rapid isolation and business continuity, and justify how your approach minimizes spread while maintaining network availability.

3. Firewall Rule Optimization under Performance Constraints

A company firewall currently uses 10,000+ rules, causing performance degradation. Constraints include:

- Throughput requirement: ≥ 800 Mbps
- Rule processing time must be minimized
- Must maintain strict access control policies
- No additional hardware allowed

Task:

Analyze inefficiencies in rule processing. Propose optimization techniques (rule aggregation, prioritization, redundancy elimination). Justify how your optimized rule set improves performance while maintaining security.

4. IPS Deployment under Resource and Accuracy Constraints

An organization wants to deploy an Intrusion Prevention System (IPS) with the following constraints:

- Must block $\geq 90\%$ of known attacks
- False positive rate $\leq 3\%$
- Available memory for signatures: 1 GB
- Network speed: 500 Mbps

Task:

Design an IPS configuration strategy (signature selection, traffic filtering, adaptive updates). Evaluate trade-offs between resource usage and detection capability, and justify how your solution satisfies all constraints.

5. Malware Detection using Behavioral Analysis under System Constraints

A company plans to implement behavioral-based malware detection with the following constraints:

- Endpoint CPU usage must remain $\leq 15\%$
- Detection must occur within 10 seconds of execution
- Must detect unknown (zero-day) threats
- Limited storage for logs and analysis

Task:

Design a lightweight behavioral detection system. Analyze trade-offs between detection speed, accuracy, and resource consumption, and justify how your approach effectively identifies malware within the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant	Identifies most constraints	Partial understanding; misses	Fails to identify constraints correctly

		constraints accurately	with minor gaps	key constraints	
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Question 1 – Intrusion Detection Deployment under Real-Time Constraints

Question

A financial organization plans to deploy an Intrusion Detection System (IDS) with the following constraints: it must analyze traffic in real time with latency ≤ 5 ms per packet; detection accuracy must be $\geq 96\%$; false-positive rate must be $\leq 1.5\%$; network traffic load is 150,000 packets per second; and CPU resources are limited because they are shared with other services. Design an optimized IDS configuration using signature-based, anomaly-based, or hybrid detection. Analyze the trade-offs between detection accuracy and system performance, and justify how the proposed solution meets all constraints.

Answer

1. Problem Analysis

Constraint	Requirement	Implication
Latency	≤ 5 ms/packet	Inspection must be lightweight and fast.
Accuracy	$\geq 96\%$	Detection must cover known and suspicious activity effectively.
False positives	$\leq 1.5\%$	Rules and anomaly thresholds must be carefully tuned.
Traffic load	150,000 packets/sec	The IDS must process high-volume traffic

		efficiently.
CPU	Limited/shared	Avoid expensive deep inspection of every packet.

2. Recommended Configuration – Hybrid IDS

A hybrid IDS is the most suitable choice. Signature-based detection provides fast identification of known attacks, while anomaly-based detection adds the ability to identify unusual or previously unseen behavior. To satisfy the CPU and latency constraints, anomaly analysis should be applied selectively to suspicious traffic rather than to every packet.

3. Configuration Steps

1. Deploy high-confidence signatures for known malware, exploits, and attack patterns.
2. Use a fast traffic-filtering stage to reduce unnecessary inspection.
3. Send suspicious or high-risk traffic to a lightweight anomaly-analysis stage.
4. Tune signatures and anomaly thresholds using normal organizational traffic.
5. Use efficient packet queues or parallel processing where supported.
6. Forward only important alerts to centralized monitoring to reduce processing overhead.

4. Trade-Off Analysis

Method	Advantages	Limitations
Signature-based	Fast; low overhead; strong for known attacks.	Weak against unknown/zero-day

		behavior.
Anomaly-based	Can detect unusual and unknown behavior.	May increase CPU usage and false positives.
Hybrid	Balances known-threat speed with behavioral detection.	Requires careful tuning and policy management.

5. Example

Normal financial transaction traffic can follow the low-cost inspection path. Packets showing a known exploit signature or unusual protocol behavior can be escalated to deeper analysis. This concentrates CPU resources on suspicious traffic while keeping ordinary traffic processing fast.

6. Verification

- Measure latency at normal and peak traffic.
- Replay known attack traffic and calculate detection accuracy.
- Replay benign traffic and calculate false-positive rate.
- Stress-test the IDS at 150,000 packets per second.
- Monitor shared CPU utilization during the test.

Final

Answer

Use a lightweight hybrid IDS with fast signature detection, selective anomaly analysis, traffic filtering, and efficient packet processing. This design provides the best balance for targeting ≤ 5 ms latency, $\geq 96\%$ detection accuracy, $\leq 1.5\%$ false positives, and 150,000 packets/sec under limited CPU resources.

Question 2 – Worm Containment Strategy under Network Availability Constraints

Question

An enterprise network faces rapid worm propagation. Network downtime must be $\leq 1\%$; detection and containment must occur within 2 minutes; systems are distributed across multiple locations; and centralized control is limited. Design a containment strategy using firewalls, IDS/IPS, and segmentation. Evaluate the trade-offs between rapid isolation and business continuity, and justify how the approach minimizes spread while maintaining network availability.

Answer

1. Problem Analysis

Constraint	Requirement	Design Response
Downtime	$\leq 1\%$	Avoid shutting down the whole enterprise.
Containment	≤ 2 minutes	Use continuous monitoring and automated response.
Locations	Multiple	Deploy local security controls.
Central control	Limited	Use decentralized and automated containment.

2. Recommended Architecture

Use a combination of distributed IDS/IPS sensors, firewalls, VLAN/network segmentation, and automated selective isolation. The network should be divided into security zones so an infected host or segment can be isolated without disconnecting healthy systems.

3. Containment Procedure

7. Continuously monitor for worm signatures, scanning, abnormal connection rates, and lateral movement.
8. Generate a high-priority alert when suspicious activity crosses a defined confidence threshold.
9. Automatically block confirmed malicious flows using IPS or firewall controls.
10. Quarantine the infected host or VLAN instead of shutting down the complete network.
11. Notify the relevant local and central security personnel.
12. Check neighboring segments and restrict unnecessary inter-segment communication.
13. Restore connectivity after remediation and verification.

4. Trade-Off

Complete network isolation may provide rapid containment but can create unacceptable business downtime. Selective isolation is slower to design and manage, but it preserves unaffected services and is better aligned with the $\leq 1\%$ availability requirement.

5. Practical Scenario

If one branch begins scanning hundreds of internal systems, the local IPS can block the suspicious traffic and isolate the affected branch VLAN. Other branches remain operational while the infected systems are investigated and cleaned.

Final

Answer

Deploy distributed IDS/IPS monitoring with firewalls, network segmentation, and automated selective isolation. This minimizes worm propagation and supports containment within 2 minutes while keeping unaffected network services available and limiting downtime to the required level.

Question 3 – Firewall Rule Optimization under Performance Constraints

Question

A company firewall currently uses 10,000+ rules, causing performance degradation. The constraints are: throughput $\geq$ 800 Mbps; rule processing time must be minimized; strict access-control policies must be maintained; and no additional hardware is allowed. Analyze inefficiencies in rule processing and propose optimization techniques including rule aggregation, prioritization, and redundancy elimination. Justify how the optimized rule set improves performance while maintaining security.

Answer

1. Problem Analysis

- Duplicate rules increase the size of the rule set.
- Shadowed or unreachable rules add unnecessary configuration complexity.
- Obsolete rules may remain after applications or users are removed.

- Poor rule ordering can increase average processing time.
- Repeated addresses and services can be grouped into reusable objects.

2. Optimization Techniques

Technique	Implementation	Benefit
Rule aggregation	Combine compatible addresses/services into groups.	Reduces number of rules.
Redundancy elimination	Remove duplicate, shadowed, and obsolete rules.	Reduces processing and complexity.
Prioritization	Place frequently matched rules earlier.	Reduces average lookup time.
Object groups	Use reusable network/service objects.	Simplifies policy management.
Default deny	Allow only required traffic; deny remaining traffic.	Maintains strong access control.

3. Example

Instead of maintaining separate rules for Host A, Host B, and Host C when they all require the same access to a web server, a compatible client group can be created and referenced by a single policy. The grouping must be checked carefully so that it does not unintentionally expand access.

4. Implementation Steps

14. Back up the current firewall configuration.
15. Inventory every rule and identify its business owner and purpose.
16. Identify duplicates, shadowed rules, unused objects, and expired policies.
17. Aggregate compatible rules and create object groups.
18. Reorder frequently matched rules while preserving policy semantics.
19. Test the optimized rule set before production deployment.
20. Deploy in stages and measure throughput and processing time.
21. Monitor denied traffic and application behavior after deployment.

5. Validation

The optimized firewall should be tested under representative traffic and peak load. Throughput must remain at or above 800 Mbps, while access-control tests confirm that authorized traffic remains available and unauthorized traffic remains blocked.

Final	Answer
Optimize the firewall using rule aggregation, prioritization, redundancy/shadow-rule elimination, object groups, and a documented default-deny policy. The result is a smaller and more efficient rule base that can improve processing performance without requiring additional hardware, while preserving strict access control.	

Question 4 – IPS Deployment under Resource and Accuracy Constraints

Question

An organization wants to deploy an Intrusion Prevention System (IPS) with the following constraints: it must block $\geq 90\%$ of known attacks; false-positive rate

must be $\leq 3\%$; available memory for signatures is 1 GB; and network speed is 500 Mbps. Design an IPS configuration strategy involving signature selection, traffic filtering, and adaptive updates. Evaluate trade-offs between resource usage and detection capability, and justify how the configuration satisfies the constraints.

Answer

1. Constraint Analysis

Constraint	Target	Solution
Known attack blocking	$\geq 90\%$	Prioritize high-value, high-confidence signatures.
False positives	$\leq 3\%$	Tune thresholds and signatures.
Signature memory	≤ 1 GB	Remove irrelevant, duplicate, and obsolete signatures.
Network speed	500 Mbps	Use efficient filtering and inspection.

2. Signature Selection

- Prioritize signatures for vulnerabilities relevant to the organization.
- Keep high-severity and actively exploited attack signatures.
- Remove duplicate or irrelevant signatures.
- Review false-positive statistics regularly.

- Use adaptive updates to replace low-value signatures with newly important ones.

3. Traffic Filtering

Traffic filtering should reduce unnecessary work before full signature inspection. Protocol-aware filtering and high-confidence policies can direct only relevant traffic to deeper inspection while preserving protection for critical services.

4. Trade-Off Analysis

Loading every available signature may improve theoretical coverage but can exceed the 1 GB memory limit and increase processing overhead. Loading too few signatures can reduce detection coverage. A risk-based signature portfolio is therefore the best compromise.

5. Verification

22. Measure signature memory usage after updates.
23. Test representative benign traffic and calculate false-positive rate.
24. Replay known attacks and estimate blocking coverage.
25. Measure throughput at 500 Mbps.
26. Tune or replace signatures based on observed results.

Final

Answer

Deploy a memory-aware IPS using relevant high-priority signatures, traffic filtering, signature tuning, and adaptive updates. The configuration should keep signature memory within 1 GB while targeting $\geq 90\%$ blocking of known attacks, $\leq 3\%$ false positives, and effective operation at 500 Mbps.

Question 5 – Malware Detection using Behavioral Analysis under System Constraints

Question

A company plans to implement behavioral-based malware detection with the following constraints: endpoint CPU usage must remain $\leq 15\%$; detection must occur within 10 seconds of execution; unknown or zero-day threats must be detected; and storage for logs and analysis is limited. Design a lightweight behavioral detection system. Analyze trade-offs between detection speed, accuracy, and resource consumption, and justify how the approach identifies malware within the given constraints.

Answer

1. Constraint Analysis

Constraint	Target	Design Response
CPU usage	$\leq 15\%$	Use lightweight continuous event monitoring.
Detection time	≤ 10 seconds	Use near-real-time risk scoring.
Unknown threats	Must be detected	Monitor suspicious behavior rather than signatures alone.
Storage	Limited	Store high-value events and summaries.

2. Behavioral Indicators

Behavior	Example	Security Meaning
Mass file modification	Many files changed rapidly.	Possible ransomware-like activity.
Process spawning	Unexpected child-process chains.	Possible exploit/script execution.
Privilege changes	Unexpected elevation.	Possible privilege escalation.
Network anomalies	Unusual outbound connections.	Possible command-and-control behavior.
Persistence changes	Unexpected startup/task changes.	Possible malware persistence.

3. Proposed Detection Workflow

27. Monitor high-value system and process events.
28. Extract lightweight behavioral features.
29. Calculate a risk score from suspicious indicators.
30. Continue normal monitoring when the score is low.
31. Generate an alert and apply the configured response when the score becomes high.
32. Preserve only important evidence to control storage use.

4. Illustrative Risk Model

An illustrative risk score can be represented as: $\text{Risk Score} = \sum (\text{behavior weight} \times \text{observed indicator})$. For example, an unexpected process alone may have low risk,

while an unexpected process combined with rapid file modification and an unusual outbound connection may produce a high score. Actual weights and thresholds must be calibrated using organizational baseline data.

5. Trade-Off Analysis

Deep continuous behavioral analysis can improve visibility but may consume excessive CPU and storage. A lightweight two-stage model is preferable: continuous low-cost monitoring followed by deeper analysis only when suspicious indicators appear.

6. Practical Example

If an unknown program starts, creates several unusual child processes, modifies a large number of files within seconds, and initiates an unexpected external connection, the combined behavioral indicators can raise the risk score and trigger an alert or isolation action even when the program has no known malware signature.

Final

Answer

Implement a lightweight behavioral detection system using real-time event monitoring, feature extraction, risk scoring, and selective deep analysis. This design minimizes CPU and storage consumption while improving the ability to identify previously unknown malicious behavior within the 10-second detection requirement.

Overall Summary

Question	Recommended Solution	Main Constraint
Q1	Hybrid IDS with selective anomaly analysis	≤ 5 ms latency; $\geq 96\%$ accuracy; $\leq 1.5\%$ false positives
Q2	Distributed IDS/IPS + segmentation + selective isolation	$\leq 1\%$ downtime; ≤ 2 min containment
Q3	Firewall rule aggregation, prioritization, redundancy elimination	≥ 800 Mbps throughput
Q4	Memory-aware IPS with high-value signatures and adaptive updates	$\geq 90\%$ blocking; $\leq 3\%$ false positives; ≤ 1 GB
Q5	Lightweight behavioral malware detection	$\leq 15\%$ CPU; ≤ 10 sec detection

Across all five questions, the preferred designs do not simply maximize security inspection. They optimize security controls around measurable constraints, preserving performance and availability while maintaining appropriate detection and prevention capability.

Glossary

Term	Definition
IDS	Intrusion Detection System that monitors activity for possible attacks.
IPS	Intrusion Prevention System that can detect and block malicious activity.
Signature	Known pattern associated with a threat or attack.
Anomaly detection	Detection based on deviation from expected behavior.
False positive	Benign activity incorrectly identified as malicious.
Network segmentation	Division of a network into controlled security zones.
Throughput	Amount of traffic processed over a period of time.
Latency	Time delay associated with processing.
Zero-day	Previously unknown or insufficiently covered vulnerability/attack.

Behavioral analysis	Detection based on observed system or process behavior.
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